

0.1 Section 6.5 - Stars and Bars

This section will just cover one idea, which is **Stars and Bars**

Let's start with an example

Example 1

Omar buys 9 icecreams. If the flavors available are chocolate and vanilla, how many possible orders can Omar make? An order in the sense like 5 chocolate 4 vanilla or 4 vanilla 5 chocolate

Let's brute force this first

Chocolate	Vanilla
0	9
1	8
2	7
3	6
4	5
5	4
6	3
7	2
8	1
9	1

So there are 10 orders possible.

Let's do this analytically.

We are essentially solving

$$a + b = 9$$

Where a and b are whole numbers (includes 0).

Thinking of this as symbols, we can write

$$\underbrace{\star\star\star\star | \star\star\star\star\star}_{10 \text{ symbols}}$$

Where the bar represents the divider between chocolate and vanilla and the stars represent the number of ice creams for each flavor.

We basically just have to count how many **arrangements** for the bar can be made. This is just a combination.

$$\binom{10}{1} = \frac{10!}{9! \cdot 1!} = \boxed{10}$$

Now that we have this example done, we can form the actual rule

Definition 1 (Stars and Bars)

If you have n slots with k options to choose from for each slot and

1. Order does not matter
2. All n slots must be filled
3. An option can be used 0 times

Then there are

$$\binom{n+k-1}{k-1} = \frac{(n+k-1)!}{n! \cdot (k-1)!}$$

Possibilities

Another way of thinking of this is stars and bars

Let

s : Stars, num slots
 b : Bars, options - 1

Then the combinations is

$$\binom{s+b}{b} = \frac{(s+b)!}{s! \cdot b!}$$

You are essentially solving for

$$x_1 + x_2 + \dots + x_k = n$$

Where

$$x_1, x_2, \dots, x_k \geq 0$$

In general, if the options must be used at least c times such that

$$x_1, x_2, \dots, x_k \geq c$$

Then the possibilities is just

$$\binom{n + k(1 - c) - 1}{k - 1}$$

For instance, if

$$x_1, x_2, \dots, x_k \geq 1$$

Possibilities are

$$\binom{n - 1}{k - 1}$$